Healthcare Predictive Analysis using EHR Data

Introduction

- Predictive healthcare analytics helps identify disease risks using patient medical records.
- This project uses anonymized Electronic Health Record (EHR) data for heart disease prediction.
- Machine Learning techniques were applied to analyze healthcare biomarkers and patient health patterns.
- Exploratory Data Analysis (EDA) and predictive modeling were performed to improve disease detection.
- The project aims to support early diagnosis and healthcare decision-making systems.

Business Problem

- Delayed disease detection increases patient risk and treatment costs.
- Manual analysis of healthcare records is time-consuming and inefficient.
- Healthcare organizations require predictive systems for early risk identification.
- Accurate disease prediction can improve clinical decision-making and patient outcomes.
- The project focuses on building a reliable and privacy-conscious healthcare prediction system.

Objectives

- Analyze Electronic Health Record (EHR) data for heart disease prediction
- Perform healthcare data preprocessing and feature engineering
- Conduct Exploratory Data Analysis (EDA)
- Build machine learning classification models
- Compare Logistic Regression and Random Forest algorithms
- Handle class imbalance using SMOT
- Evaluate models using healthcare-focused metrics
- Improve early disease detection and healthcare decision support

Dataset Overview

The dataset contains anonymized healthcare records associated with heart disease prediction. The dataset includes patient medical attributes such as:

- Age
- Sex
- Chest Pain Type
- Resting Blood Pressure
- Cholesterol
- Fasting Blood Sugar

- Resting ECG
- Maximum Heart Rate
- Exercise-Induced Angina
- ST Depression
- Heart Disease Status

Target Variable:

- 0 = No Heart Disease
- 1 = Heart Disease Present

Data Preprocessing

Data preprocessing was performed to improve dataset quality and prepare the healthcare records for machine learning implementation.

The preprocessing phase included:

- Handling missing values
- Removing duplicate records
- Data cleaning
- Label encoding categorical variables
- Feature preparation for predictive modeling

Categorical healthcare variables were transformed into numerical values for machine learning compatibility.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to identify healthcare trends, patient distributions, and relationships between medical biomarkers and heart disease occurrence.

The following visualizations were analyzed:

Heart Disease Distribution**:** This graph showed the distribution of healthy and diseased patients in the dataset.

Age Distribution**:** The analysis demonstrated that middle-aged and older patients had a higher frequency of heart disease occurrence.

Gender Distribution**:** Male patients appeared more frequently in the heart disease category compared to female patients.

Cholesterol Distribution**:** Higher cholesterol levels were associated with increased cardiovascular risk.

Blood Pressure Distribution**:** Elevated resting blood pressure levels were observed among several patients diagnosed with heart disease.

Maximum Heart Rate Distribution**:** The graph analyzed variations in patient heart rates and their relationship with disease occurrence.

Age vs Heart Disease**:** Older age groups showed a higher tendency toward heart disease prediction.

Correlation Heatmap**:** The correlation heatmap identified relationships between healthcare biomarkers and the target variable.

Chest Pain Type vs Heart Disease**:** Certain chest pain categories, especially asymptomatic chest pain, demonstrated stronger association with heart disease.

Exercise-Induced Angina vs Heart Disease**:** Patients experiencing exercise-induced angina showed higher heart disease frequency.

Fasting Blood Sugar vs Heart Disease**:** Elevated fasting blood sugar levels indicated increased cardiovascular risk.

Resting ECG vs Heart Disease**:** Abnormal ECG patterns were more frequently observed among heart disease patients.

Machine Learning Implementation

Train-Test Split

The dataset was divided into training and testing subsets using an 80:20 ratio to evaluate model performance on unseen healthcare records.

Handling Class Imbalance using SMOT

SMOT (Synthetic Minority Oversampling Technique) was applied to address class imbalance in the healthcare dataset and improve predictive performance for minority disease classes.

Logistic Regression Model

Logistic Regression was implemented as the baseline machine learning classification model for heart disease prediction.

The model was trained using scaled healthcare features and evaluated using:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC

The model achieved strong classification performance while maintaining interpretability for healthcare analytics.

Random Forest Model

Random Forest was implemented as an ensemble machine learning algorithm to improve prediction accuracy and reduce overfitting.

The model combined multiple decision trees to classify patients based on healthcare biomarkers and clinical attributes.

Random Forest demonstrated higher predictive capability compared to Logistic Regression.

Model Evaluation

Classification Report

Both models were evaluated using:

- Accuracy
- Precision

- Recall
- F1-score

Special attention was given to Recall because false negative predictions in healthcare can delay treatment and increase patient risk.

Confusion Matrix

Confusion matrices were generated to visualize correctly and incorrectly classified patient records.

ROC-AUC Curve

ROC curves and AUC scores were used to evaluate the classification capability of both machine learning models.

The Random Forest model achieved stronger ROC-AUC performance, indicating improved disease prediction capability.

Feature Importance Analysis

Feature importance analysis identified the most influential healthcare biomarkers contributing to heart disease prediction.

Important predictive features included:

- Chest Pain Type
- Maximum Heart Rate
- Cholesterol
- Resting ECG
- Exercise-Induced Angina

Results

Among the implemented machine learning algorithms, Random Forest demonstrated better classification performance compared to Logistic Regression.

Key outcomes:

- Improved disease prediction accuracy
- Better Recall performance
- Enhanced patient risk classification
- Effective handling of class imbalance using SMOTE

The project successfully demonstrated the effectiveness of machine learning in predictive healthcare analytics.

Conclusion

- The project successfully implemented predictive healthcare analytics using EHR data.
- Data preprocessing, EDA, and machine learning techniques were applied for heart disease prediction.
- Logistic Regression and Random Forest models were developed and evaluated.
- Random Forest demonstrated better predictive performance and disease classification capability.

- The project highlights the importance of machine learning in early healthcare risk detection and clinical decision support systems.

Future Scope

- Integration with real-time healthcare systems
- Deployment using Streamlit or Flask
- Deep Learning implementation
- Real-time patient risk monitoring dashboards
- Advanced healthcare analytics and visualization systems